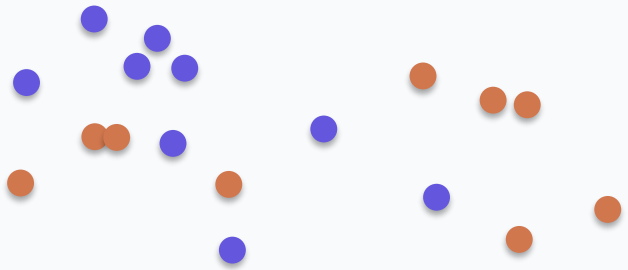
Probably Approximately Correct (PAC) Learning

$$P(\text{error}(h) \leq \varepsilon) \geq 1 - \delta$$

Learning

TRAINING DATA

Examples the learner sees



UNSEEN DATA

must predict



1. Distribution → **2. Sample** → **3. Algorithm** → **4. Model**

Unknown D → S of m examples → A(S) → h = A(S)

True error: $\text{error}_D(h) = P_{(x,y)\sim D}[\, h(x) \neq y \,]$

Probability of misclassifying a fresh example from D.

One formula for two tolerances

$$P_s(\text{error}_D(h) \leq \varepsilon) \geq 1 - \delta$$

h	$\text{error}_D(h)$	ε	δ
hypothesis	True error on new data	Maximum Acceptable error	Maximum chance of failure

$\varepsilon = 0.10,\ \delta = 0.05$

Consider a learner with a valid PAC guarantee at these tolerances.

$\varepsilon = 0.10$

True error at most 10%

$\delta = 0.05$

Confidence at least 95%

95% of training runs produce a model with ≤ 10% true error.

Imagine repeating the training process

Each independent run **draws a new sample** and **learns a new model**.

Different sample -> different model

The PAC probability is over possible training runs

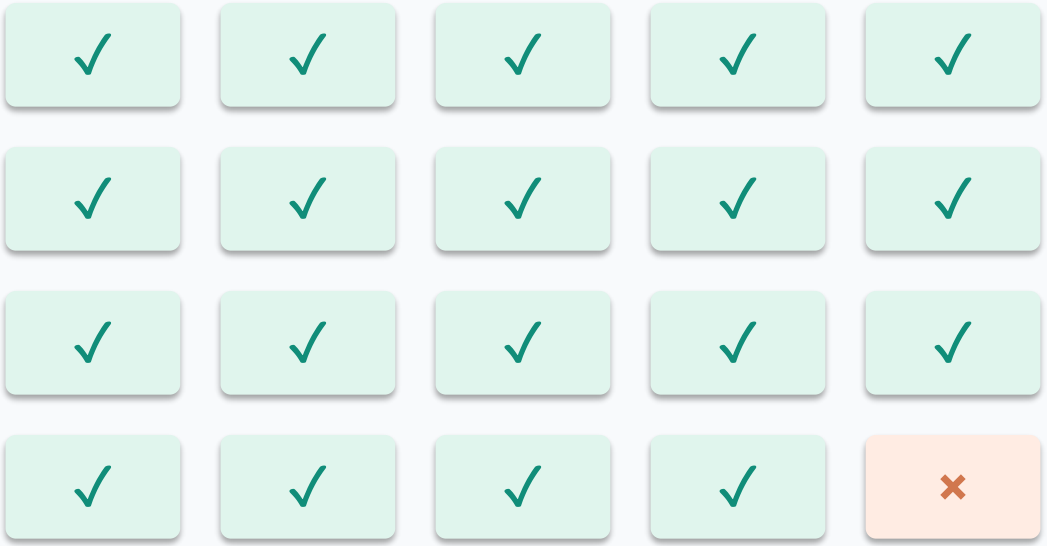


Illustration: 19 successful runs / 1 unsuccessful

What happens if we demand more?

Smaller ε or smaller δ calls for more examples.

Accuracy

ε: 0.20 → 0.05

We tolerate fewer errors.

Confidence

δ: 0.20 → 0.01

We tolerate fewer **failed runs.**

Better guarantees can demand more training data.

A classic guarantee

Realizable case: the correct hypothesis belongs to finite H.

$$m \geq (\ln |H| + \ln(1/\delta)) / \varepsilon$$

m = training-set size • |H| = number of hypotheses

EXAMPLE

$|H| = 1{,}000$

$\varepsilon = 0.10, \delta = 0.05$

$$m \geq 100$$

Assumptions: finite H, realizability, and a hypothesis fitting the training sample.

PAC: reliable learning with error

Approximately correct	Population error is at most ε.
Probably	This happens with probability at least $1 - \delta$.
Generalization	Guarantees concern unseen examples from D.

PAC guarantees require clear assumptions about data, hypotheses and learner